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Vehicle management system

Abstract

In a case where a travel permission request regarding a queuing zone for a first unmanned vehicle stopped in a queuing zone is received, on the basis of a time length taken for the first unmanned vehicle arriving at a deceleration start position and a time length taken for a second unmanned vehicle positioned in a loading zone going out of the loading zone from reception of a loading completion notification, a travel start time difference after which the second unmanned vehicle goes out of the loading zone before the first unmanned vehicle arrives at a deceleration start position is calculated, a travel start time of the first unmanned vehicle is calculated on the basis of a reception time of the loading completion notification and the travel start time difference, and a travel permission regarding the queuing zone for the first unmanned vehicle is output after the travel start time has come. As a result, it is possible to shorten a time length required for replacement of unmanned vehicles at a loading position and enhance productivity.

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Background/Summary**TECHNICAL FIELD**

(1) The present invention relates to a vehicle management system.

BACKGROUND ART

(2) At strip mines or the like, vehicle management systems including dump trucks that autonomously travel without operators getting in them (i.e., unmanned vehicles), and a traffic management station that communicates with the unmanned vehicles via a wireless communication line have been used.

(3) As a technology related to vehicle management systems, for example, one described in Patent Document 1 has been known. Patent Document 1 discloses a vehicle travel system that generates a vehicle travel route where a vehicle travels from an entrance point of a loading site to a loading point where there is a loading machine and that causes the vehicle to travel along the generated travel route, the vehicle travel system including travel route generating means that generates a travel route from the entrance point to the loading point through a queuing point near the loading point on the basis of positional information regarding the loading point and positional information regarding the entrance point, first travel control means that causes the vehicle to travel from the entrance point to the queuing point along the travel route generated by the travel route generating means, on the basis of information regarding the travel route, queuing means that causes the vehicle to queue at the queuing point until a permission is obtained from the loading machine, partial travel route generating means that generates a partial travel route from the queuing point to the loading point at a different position changed due to a positional change, on the basis of positional information regarding the loading point at the different position changed due to the positional change and positional information regarding the queuing point on the travel route before the positional change of the loading point in a case where, while the vehicle is queuing at the queuing point or traveling from the entrance point to the queuing point, there is an instruction for the positional change of the loading point from the loading machine to a traffic managing device or/and the vehicle, and second travel control means that causes the vehicle to travel from the queuing point to the loading point along the travel route generated by the travel route generating means, on the basis of the information regarding the travel route, in a case where, while the vehicle is queuing at the queuing point and traveling from the entrance point to the queuing point, there is not the instruction for the positional change of the loading point from the loading machine to the

traffic managing device or/and the vehicle, and causes the vehicle to travel from the queuing point to the loading point at the different position changed due to the positional change along the partial travel route generated by the partial travel route generating means, on the basis of information regarding the partial travel route, in a case where, while the vehicle is queuing at the queuing point or traveling from the entrance point to the queuing point, there is the instruction for the positional change of the loading point.

PRIOR ART DOCUMENT

Patent Document

(4) Patent Document 1: WO2011/090093

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(5) In the conventional technology described above, on the travel route of the unmanned vehicle, the queuing point like a switchback point is set near the loading point, as a point where the vehicle should queue until a permission is obtained from the loading machine, and the unmanned vehicle is caused to keep traveling to the queuing point, that is, to a position near the loading machine, avoiding stops as much as possible, thereby aiming for enhancement of production efficiency.

(6) However, a queuing point needs to be set at a position well distanced from a loading point to prevent, for example, contact between unmanned vehicles. In particular, since very large-sized unmanned vehicles are operated at a strip mine or the like in many cases and moving distances required for action changes such as acceleration/deceleration of the unmanned vehicles are taken into consideration, a queuing point is set to a position distanced from a loading point to some extent. In addition, in a case where timings of travel start from a queuing point of unmanned vehicles are not taken into consideration as in the conventional technology described above, a time length required for replacement of unmanned vehicles at a loading point becomes long, and the productivity deteriorates undesirably.

(7) The present invention has been made in view of the problems described above, and an object thereof is to provide a vehicle management system that can shorten a time length required for replacement of unmanned vehicles at a loading position and can enhance productivity.

Means for Solving the Problems

(8) The present application includes a plurality of means for solving the problems described above, and an example thereof is a vehicle management system in which a plurality of unmanned vehicles including at least a first unmanned vehicle and a second unmanned vehicle that haul to-be-hauled objects and are capable of traveling autonomously, a loading machine that performs loading work of loading the to-be-hauled objects onto each of the plurality of unmanned vehicles, and a traffic management station that performs traffic management control of the loading machine and the plurality of unmanned vehicles are wirelessly connected to each other in a mutually communicable manner. The vehicle management system includes an unmanned vehicle controller, a loading completion notification input device, and a traffic management controller. The unmanned vehicle controller acquires vehicle body information including positional information representing positions of the plurality of unmanned vehicles at a work site and orientation information representing orientations of the plurality of unmanned vehicles at the work site, controls autonomous traveling of the plurality of unmanned vehicles on the basis of a travel permission for the plurality of unmanned vehicles, and outputs a travel permission request for requesting a travel permission, for each of the plurality of unmanned vehicles, regarding a predetermined travel zone among a plurality of travel zones included in a travel route where the plurality of unmanned vehicles travel. The loading completion notification input device outputs, according to input operation by an operator of the loading machine, a loading completion notification indicating that loading work of loading the to-be-hauled objects onto corresponding one of the unmanned vehicles that is stopped at a loading position preset on the travel route as a position where the loading work is performed by the loading machine for the unmanned vehicles has been completed. The traffic

management controller sets a travel route where the plurality of unmanned vehicles travel, on the basis of a work schedule predetermined for the work site and map information including information regarding positions and velocity limits of a plurality of travel routes and that outputs a travel permission regarding a predetermined travel zone of the travel route set for the plurality of unmanned vehicles, on the basis of a travel permission request from the unmanned vehicle controller. The traffic management controller is configured to, in a case where a loading zone which is a travel zone including the loading position and in which only one unmanned vehicle is allowed to enter at once and a queuing zone which is a travel zone adjacent to the loading zone and in which an unmanned vehicle heading toward the loading zone queues are set and where a travel permission request regarding the queuing zone by the first unmanned vehicle stopped in the queuing zone is received, calculate such a travel start time difference between the first unmanned vehicle and the second unmanned vehicle positioned in the loading zone that the second unmanned vehicle goes out of the loading zone a predetermined spare time length before the first unmanned vehicle arrives at a deceleration start position where the first unmanned vehicle should start decelerating for stopping before entering the loading zone, on the basis of a time length for arriving at the deceleration start position that is a time length taken for the first unmanned vehicle arriving at the deceleration start position after starting traveling, and on the basis of a time length for arriving at a loading zone releasing position that is a time length taken for the second unmanned vehicle going out of the loading zone after starting traveling in response to reception of the loading completion notification to the second unmanned vehicle, calculate a travel start time of the first unmanned vehicle on the basis of a reception time of the loading completion notification to the second unmanned vehicle and the travel start time difference, and output a travel permission regarding the queuing zone for the first unmanned vehicle after the travel start time has come.

Advantages of the Invention

(9) According to the present invention, it is possible to shorten a time length required for replacement of unmanned vehicles at a loading position and enhance productivity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a figure depicting the overall configuration of a vehicle management system.
- (2) FIG. 2 is a figure depicting the external appearance of a hydraulic excavator depicted as an example of loading machines.
- (3) FIG. 3 is a figure depicting an overview of an operator's seat installed in an operation room.
- (4) FIG. 4 is a side view schematically depicting the external appearance of a dump truck depicted as an example of unmanned vehicles.
- (5) FIG. 5 is a functional block diagram depicting the vehicle management system.
- (6) FIG. 6 is a figure depicting an example of a table of vehicle allocation management information.
- (7) FIG. 7 is a figure depicting an example of a table of traffic management information stored on a traffic management information storage section.
- (8) FIG. 8 is a figure depicting an example of a travel route set for a work area where loading work is performed.
- (9) FIG. 9 is a flowchart depicting the processing content of a process of giving permission of travel start, the process being performed by a traffic management controller.
- (10) FIG. 10 is a flowchart depicting the processing content of an estimation process of estimating a time length for releasing a loading zone.
- (11) FIG. 11 is a figure for explaining a processing principle of the estimation process of estimating the time length for releasing the loading zone.

- (12) FIG. 12 is a flowchart depicting the processing content of an estimation process of estimating a time length for arriving at a deceleration start position.
- (13) FIG. 13 is a figure for explaining a processing principle of the estimation process of estimating the time length for arriving at the deceleration start position.
- (14) FIG. 14 is a flowchart depicting the processing content of a travel start time determination process.
- (15) FIG. 15 is a figure for explaining a processing principle of the travel start time determination process.
- (16) FIG. 16 is a figure for explaining the processing principle of the travel start time determination process.
- (17) FIG. 17 is a figure for explaining the processing principle of the travel start time determination process.
- (18) FIG. 18 is a figure depicting a state of vehicle management.
- (19) FIG. 19 is a figure depicting changes of the position and velocity of an unmanned vehicle.
- (20) FIG. 20 is a figure depicting a state of vehicle management in a conventional technology depicted as a comparative example.
- (21) FIG. 21 is a figure depicting changes of the position and velocity of an unmanned vehicle.
- (22) FIG. 22 is a flowchart depicting the processing content of the process of giving permission of travel start, the process being performed by the traffic management controller according to a second embodiment.
- (23) FIG. 23 is a figure for explaining a calculation principle of an optimum traveling velocity according to the second embodiment.
- (24) FIG. 24 is a figure for explaining a calculation principle of the optimum traveling velocity according to a modification example of the second embodiment.
- (25) FIG. 25 is a flowchart depicting the processing content of the process of giving permission of travel start, the process being performed by the traffic management controller according to a third embodiment.
- (26) FIG. 26 is a figure for explaining a processing principle of travel permission output according to a fourth embodiment.
- (27) FIG. 27 is a figure depicting a state of vehicle management according to the fourth embodiment.
- (28) FIG. 28 is a figure depicting an example of a travel route according to a fifth embodiment.

MODES FOR CARRYING OUT THE INVENTION

(29) Hereinbelow, embodiments of the present invention are explained with reference to the figures. Note that, whereas dump trucks as an example of haulage vehicles which are management subjects of a vehicle management system and hydraulic excavators as an example of loading machines which are management subjects of the vehicle management system are depicted and explained in the present embodiments, the present invention can be applied also to a vehicle management system whose management subjects are other haulage vehicles or loading machines.

First Embodiment

(30) A first embodiment of the present invention is explained with reference to FIG. 1 to FIG. 21.

(31) FIG. 1 is a figure depicting the overall configuration of the vehicle management system according to the present embodiment.

(32) As depicted in FIG. 1, a vehicle management system **100** is a system used for work sites such as strip mines, and includes one or more loading machines **10** that perform excavation work and loading work, one or more unmanned vehicles **20** (an unmanned vehicle **20-1** and an unmanned vehicle **20-2**) that transport to-be-hauled objects such as earth and sand loaded from the loading machines **10**, and a traffic management station **30** that performs vehicle allocation management and traffic management of the unmanned vehicles **20**. The loading machines **10**, the unmanned vehicles **20**, and the traffic management station **30** are configured to be able to communicate with each other

by a wireless communication line **40**. Specifically, a plurality of wireless base stations **41** are installed at a strip mine or the like, and the loading machines **10**, the unmanned vehicles **20**, and the traffic management station **30** perform transmission and reception to and from each other via the wireless base stations **41**.

(33) In a control scheme used as a traffic management scheme of the traffic management station **30** in the present embodiment, exclusive travel permissions regarding partial zones (travel zones) of a transportation route **60** divided by nodes on map data representing the transportation route **60** are given on the basis of the positions of the respective unmanned vehicles **20** (what is generally called a travel permitted zone control scheme). In the travel permitted zone control scheme, for example, in a case where a subject unmanned vehicle **20** requests a travel permission regarding a travel zone ahead of the subject unmanned vehicle **20** and, at that time, another unmanned vehicle has been given a travel permission regarding the travel zone ahead regarding which the subject unmanned vehicle **20** requests the travel permission or the travel zone ahead is set as an entry prohibited zone, the subject unmanned vehicle **20** is not given the travel permission regarding the travel zone ahead. Accordingly, the subject unmanned vehicle **20** stops at, for example, the end of a currently permitted zone, and queues until the travel permission regarding the travel zone ahead is given.

(34) FIG. 2 is a figure depicting the external appearance of a hydraulic excavator depicted as an example of the loading machines. Note that the loading machines **10** are not limited to hydraulic excavators, and may be wheel loaders or the like, for example.

(35) As depicted in FIG. 2, a hydraulic excavator (loading machine) **10** includes an articulated-type front device **10A** including a vertically pivotable boom **191**, arm **192**, and bucket **193** that are linked with each other, an upper swing structure **10B**, and a lower travel structure **10C**. In addition, an operation room **197** where an operator gets in is arranged at the upper front of the upper swing structure **10B**.

(36) The base end of the boom **191** of the front device **10A** is supported pivotably by a front section of the upper swing structure **10B**, one end of the arm **192** is supported pivotably by an end section (front end) which is different from the base end of the boom **191**, and the bucket **193** is supported pivotably by the other end of the arm **192**. The boom **191**, the arm **192**, the bucket **193**, the upper swing structure **10B**, and the lower travel structure **10C** are driven by a boom cylinder **194**, an arm cylinder **195**, a bucket cylinder **196**, an undepicted swing motor, and undepicted left and right travel motors, respectively. Hereinafter, the boom cylinder **194**, the arm cylinder **195**, the bucket cylinder **196**, the swing motor, and the left and right travel motors are collectively referred to as a vehicle body drive system **180** (see FIG. 5 below) in some cases.

(37) FIG. 3 is a figure depicting an overview of an operator's seat installed in the operation room.

(38) As depicted in FIG. 3, operation levers **171** (a right operation lever **171-1** and a left operation lever **171-2**) that output operation signals for operating the hydraulic actuators **194** to **196** or the like are provided at an operator's seat **172** installed in the operation room **197**. The operation levers **171** can be inclined forward/backward and leftward/rightward, include undepicted sensors that electrically sense lever inclination amounts, that is, lever operation amounts, which are operation signals, and output, via electric wires, the lever operation amounts sensed by the sensors. That is, operation of each of the hydraulic actuators **194** to **196** or the like is assigned to the forward/backward directions or leftward/rightward directions of the operation levers **171**. Hereinafter, the operation levers **171** including the right operation lever **171-1** and the left operation lever **171-2** are referred to as a vehicle body operation input device **171** (see FIG. 5 below) in some cases.

(39) In addition, at a top section of the operation lever **171** (e.g., the right operation lever **171-1**), a loading completion notification input device **170** is arranged as a switch that can be quickly operated at any time by, for example, the thumb of a right hand while the operation lever **171** is being gripped.

(40) According to input operation by an operator of the loading machine (hydraulic excavator **10**),

the loading completion notification input device **170** outputs a loading completion notification indicating that loading work for the unmanned vehicle **20** stopped at a loading position preset on a travel route as a position where the loading machine performs loading work of loading to-be-hauled objects onto the unmanned vehicles **20** is completed.

(41) Note that, for example, it is sufficient if the loading completion notification input device **170** is a switch or the like arranged at the operator's seat **172** such that an operator can easily operate the loading completion notification input device **170**, and need not necessarily be a switch arranged at the top section of the operation lever **171**. For example, as long as the loading completion notification input device **170** is one that allows an operator to perform input operation while performing normal operation such as loading, the loading completion notification input device **170** may be a device other than a switch, and may not be one arranged at the top section of the operation lever **171**.

(42) FIG. **4** is a side view schematically depicting the external appearance of a dump truck depicted as an example of the unmanned vehicles (haulage vehicles). Note that, in FIG. **4**, only one of a pair of left and right constituent elements such as follower wheels, driving wheels, or travel motors is depicted and given a reference character, and the other is represented only by a reference character in parentheses in the figure but is not depicted.

(43) As depicted in FIG. **4**, for example, a dump truck **20** which is an unmanned vehicle is an electrically driven dump truck, and includes, in general, a vehicle body frame **281** that extends in the forward/backward directions and that forms a support structure, a cargo bed (vessel) **285** that is arranged to extend in the forward/backward directions at a top section of the vehicle body frame **281** and that has a lower rear end section provided to the vehicle body frame **281** to be able to be inclined via a pin-joining section **285a**, a pair of follower wheels (front wheels) **282L** and **282R** that are provided on the left side and right side at the lower front of the vehicle body frame **281**, a pair of driving wheels (rear wheels) **283L** and **283R** provided on the left side and right side at the lower back of the vehicle body, an operation room **284** provided at the upper front of the vehicle body frame **281**, a fuel tank **289** provided below the vehicle body frame **281**, an engine (not depicted) that is arranged on the vehicle body frame **281** and that is driven by a fuel supplied from the fuel tank **289**, and an electric drive system having, for example, travel motors **290L** and **290R** that drive the wheels (driving wheels **283L** and **283R**) by using electric power output from an electric power generator connected to and driven by the engine. The travel motors **290L** and **290R** are housed in rotation shaft sections of the driving wheels **283L** and **283R** along with undepicted velocity reduction gears. The vehicle body frame **281** and the cargo bed **285** are connected to each other by a hoist cylinder **286**, and the cargo bed **285** is pivoted about the pin-joining section **285a** by expansion and contraction of the hoist cylinder **286**.

(44) FIG. **5** is a functional block diagram depicting the vehicle management system.

(45) As depicted in FIG. **5**, the vehicle management system **100** includes a plurality of unmanned vehicles **20** that haul to-be-hauled objects such as earth and sand and are capable of traveling autonomously, a loading machine **10** that performs loading work of loading the to-be-hauled objects onto each of the plurality of unmanned vehicles **20**, and the traffic management station **30** that performs traffic management control of the loading machine **10** and the plurality of unmanned vehicles **20**. The plurality of unmanned vehicles **20**, the loading machine **10**, and the traffic management station **30** are wirelessly connected to each other in a mutually communicable manner.

(46) An unmanned vehicle controller **200** is arranged for each of the plurality of unmanned vehicles **20**. The unmanned vehicle controller **200** acquires vehicle body information including positional information representing the position of the unmanned vehicle **20** at a work site and orientation information representing the orientation of the unmanned vehicle **20** at the work site, controls autonomous traveling of the unmanned vehicle **20** on the basis of a travel permission for the unmanned vehicle **20**, and outputs a travel permission request for requesting a travel permission regarding a predetermined travel zone among a plurality of travel zones included in a travel route

where the unmanned vehicle **20** travels.

(47) The loading completion notification input device **170** is arranged in the loading machine **10**. According to the input operation by an operator of the loading machine **10**, the loading completion notification input device **170** outputs a loading completion notification indicating that loading work for the unmanned vehicle **20** stopped at a loading position preset on a travel route as a position where the loading machine **10** performs loading work of loading to-be-hauled objects onto the unmanned vehicles **20** is completed.

(48) A traffic management controller **310** is arranged in the traffic management station **30**. The traffic management controller **310** sets a travel route where the plurality of unmanned vehicles **20** travel on the basis of a predetermined work schedule of the work site and map information including information regarding the positions and velocity limits of a plurality of travel routes, and outputs a travel permission regarding a predetermined travel zone of the travel route set for the plurality of unmanned vehicles **20**, on the basis of a travel permission request from an unmanned vehicle controller **200**.

(49) Note that one loading machine **10** and one unmanned vehicle **20** are depicted in FIG. 5, but, also in a case where there are two or more loading machines **10** and two or more unmanned vehicles **20**, they have similar configurations.

(50) (Loading Machine **10**)

(51) The loading machine **10** includes the vehicle body operation input device **171**, the vehicle body drive system **180**, the loading completion notification input device **170**, and a wireless communication device **140**.

(52) For example, the wireless communication device **140** is radio equipment for connection to the wireless communication line **40**. The wireless communication device **140** performs information transmission and reception to and from the unmanned vehicles **20** or the traffic management station **30** via the wireless communication line **40**.

(53) The vehicle body operation input devices **171** are the operation levers **171-1** and **171-2**, and are provided at the operator's seat **172**.

(54) The vehicle body drive system **180** is the boom cylinder **194**, the arm cylinder **195**, the bucket cylinder **196**, the swing motor, and the left and right travel motors, and is driven according to operation signals from the vehicle body operation input devices **171**. In addition, the vehicle body drive system **180** is provided with an undepicted inertial measurement unit (IMU), and can transmit postural information regarding the vehicle body drive system **180** or the number of times of loading obtained from the postural information to the unmanned vehicles **20** or the traffic management station **30** via the wireless communication device **140**.

(55) According to input operation by an operator of the loading machine (hydraulic excavator **10**), the loading completion notification input device **170** outputs a loading completion notification indicating that loading work for the unmanned vehicle **20** stopped at a loading position preset on a travel route as a position where the loading machine performs loading work of loading to-be-hauled objects onto the unmanned vehicles **20** is completed. The loading completion notification from the loading completion notification input device **170** is transmitted to the unmanned vehicle **20** or the traffic management station **30** via the wireless communication device **140**.

(56) The overall action of the loading machine **10** is controlled by an undepicted controller. For example, the controller of the loading machine **10** is configured by a microcomputer which is a combination of a CPU (Central Processing Unit) that executes computations, a ROM (Read Only Memory) as a secondary storage device having recorded thereon a program for the computations, and a RAM (Random Access Memory) as a transitory storage device that stores intermediate results of the computations and stores temporary control variables. The controller of the loading machine **10** controls actions of the loading machine **10** by execution of the stored program.

(57) (Unmanned Vehicles **20**)

(58) The unmanned vehicle **20** includes the unmanned vehicle controller **200**, a travel drive system

210, a position/orientation sensor **220**, a velocity sensor **230**, a loading sensor **270**, a storage device **250**, and a wireless communication device **240**.

(59) The travel drive system **210** drives traveling of the unmanned vehicle **20** on the basis of control signals of the unmanned vehicle controller **200**. For example, the travel drive system **210** includes a steering motor for changing the steering angle of the unmanned vehicle **20**, the travel motors **290L** and **290R** for causing the unmanned vehicle **20** to travel, a brake, and the like.

(60) For example, the position/orientation sensor **220** is a GPS (Global Positioning System) device, a magnetic sensor, or the like. The position/orientation sensor **220** measures the position and orientation of the subject unmanned vehicle **20**, and outputs the measured position and orientation to the unmanned vehicle controller **200**. Note that the position/orientation sensor **220** may be a combination of a GPS and an inertial measurement unit (IMU), or one that identifies the position by using radio waves from base stations installed on the ground.

(61) The loading sensor **270** measures the weight of objects loaded on the unmanned vehicle **20** (i.e., a loaded amount). The loading sensor **270** may be a weight sensor included at a seat portion of the vessel (cargo bed) **285**, or may be one that estimates the weight on the basis of the pressure of the hoist cylinder **286** that cause the vessel **285** to perform an action. The loading sensor **270** outputs the measured loaded amount to the unmanned vehicle controller **200**.

(62) The storage device **250** is a non-volatile storage medium from which information can be read out and on which information can be written. The storage device **250** has stored thereon an OS (Operating System) and various types of control programs, application program, database, and the like. In addition, a storage region as a map information storage section **251** is formed on the storage device **250**. The map information storage section **251** stores a work schedule predetermined for a work site and map information including information regarding the positions and velocity limits of a plurality of travel routes.

(63) For example, the wireless communication device **240** is radio equipment for connection to the wireless communication line **40**. The wireless communication device **240** performs information transmission and reception to and from the loading machine **10** or the traffic management station **30** via the wireless communication line **40**.

(64) For example, the unmanned vehicle controller **200** is configured by a microcomputer which is a combination of a CPU (Central Processing Unit) that executes computations, a ROM (Read Only Memory) as a secondary storage device having recorded thereon a program for the computations, and a RAM (Random Access Memory) as a transitory storage device that stores intermediate results of the computations and stores temporary control variables. The unmanned vehicle controller **200** controls actions of the unmanned vehicle **20** by execution of the stored program.

(65) The unmanned vehicle controller **200** has an autonomous travel control section **201**, a vehicle body information managing section **202**, and a travel permission requesting section **203**. The vehicle body information managing section **202** manages information regarding the position and orientation output from the position/orientation sensor **220** and information regarding the loaded amount output from the loading sensor **270**, and transmits these pieces of information to the traffic management station **30** via the wireless communication device **240**. In addition, the vehicle body information managing section **202** outputs information regarding the position, orientation and loaded amount to the autonomous travel control section **201**. Further, in a case where the vehicle body information managing section **202** has received information regarding a travel route and a travel permitted zone for the subject unmanned vehicle **20** from a traffic management control section **312** (described later) of the traffic management station **30**, the vehicle body information managing section **202** outputs the received information to the autonomous travel control section **201**.

(66) On the basis of the information regarding the position, orientation, loaded amount, travel route, and travel permitted zone output from the vehicle body information managing section **202**, the autonomous travel control section **201** generates control signals such as

acceleration/deceleration control signals or steering control signals for causing the unmanned vehicle **20** to travel such that the unmanned vehicle **20** does not deviate from the travel permitted zone while following the travel permitted route. Further, the autonomous travel control section **201** outputs these generated control signals to the travel drive system **210**.

(67) (Traffic Management Station **30**)

(68) The traffic management station **30** includes the traffic management controller **310**, a traffic management storage device **350**, and a wireless communication device **340**.

(69) The traffic management storage device **350** is a non-volatile storage medium from which information can be read out and on which information can be written. The traffic management storage device **350** has stored thereon an OS (Operating System) and various types of control programs, application program, database, and the like. Storage regions as a vehicle allocation management information storage section **351**, a traffic management information storage section **352**, and a map information storage section **353** are formed on the traffic management storage device **350**.

(70) For example, the wireless communication device **340** is radio equipment for connection to the wireless communication line **40**. The wireless communication device **340** performs information transmission and reception to and from the loading machine **10** or the unmanned vehicles **20** via the wireless communication line **40**.

(71) For example, the traffic management controller **310** is configured by a microcomputer which is a combination of a CPU (Central Processing Unit) that executes computations, a ROM (Read Only Memory) as a secondary storage device having recorded thereon a program for the computations, and a RAM (Random Access Memory) as a transitory storage device that stores intermediate results of the computations and stores temporary control variables. The traffic management controller **310** controls actions of the traffic management station **30** by execution of the stored program.

(72) The traffic management controller **310** has a vehicle allocation managing section **311**, the traffic management control section **312**, and a substitute vehicle travel-start time calculating section **313**.

(73) The vehicle allocation managing section **311** sets a travel route to a destination of the unmanned vehicles **20**. For example, in a case where the unmanned vehicle **20** is at a loading position, the vehicle allocation managing section **311** sets a travel route to a soil-dropping position. On the other hand, in a case where the unmanned vehicle **20** is at a soil-dropping position, the vehicle allocation managing section **311** sets a travel route to a loading position. Then, travel routes set by the vehicle allocation managing section **311** are stored on the vehicle allocation management information storage section **351** in, for example, a table format as vehicle allocation management information.

(74) FIG. **6** is a figure depicting an example of a table of the vehicle allocation management information.

(75) As depicted in FIG. **6**, in the vehicle allocation management information, a travel route set by the vehicle allocation managing section **311** is recorded as a target route for each vehicle ID uniquely identifying the unmanned vehicle **20**. The target route represents a route from a loading position (node_LP) to a soil-dropping position (node_DP) or from the soil-dropping position (node_DP) to the loading position (node_LP).

(76) As for target routes, map information that can be preset as travel routes like a transportation route between work areas where loading work or soil-dropping work is performed is stored on the map information storage section **353** in a format in line with the shapes of the transportation routes in advance. On the other hand, map information that cannot be preset as travel routes like a travel zone including work points such as a loading position, a queuing position, or a soil-dropping position is generated by the traffic management control section **312** when a loading position or a soil-dropping position, which is a work point, is specified, and is stored on the map information

storage section **353**. Note that only one piece or a plurality of pieces of information may be generated as the map information generated by the traffic management control section **312**. In a case where a plurality of pieces of information are generated, the vehicle allocation managing section **311** selects one of the plurality of pieces of map information when a travel route is to be set for the unmanned vehicle **20**.

(77) In addition, in a case where there is map information having been generated for a destination work area, when a travel route is to be set for the unmanned vehicle **20**, the vehicle allocation managing section **311** sets a travel route between work areas and a travel route in the work areas simultaneously. On the other hand, in a case where there is not map information having been generated for a destination work area, it is sufficient if the vehicle allocation managing section **311** first sets a travel route to the entrance point of a destination work area and, at a time point when a work point such as a loading position, a queuing position, or a soil-dropping position in the work area has been specified, sets a travel route in the work area.

(78) On the basis of information regarding traffic management stored on the traffic management information storage section **352** (hereinafter, simply referred to as “traffic management information”), the traffic management control section **312** sets each of a plurality of travel zones set for travel routes of the unmanned vehicles **20** as a travel permitted zone regarding which a travel permission is given to only any one of the plurality of unmanned vehicles **20**. That is, there is only one unmanned vehicle **20** to which a travel permission regarding each travel zone of a plurality of travel zones is given, and there can never be a case where there are a plurality of the unmanned vehicles **20** that are given a travel permission regarding one travel zone.

(79) FIG. 7 is a figure depicting an example of a table of the traffic management information stored on the traffic management information storage section.

(80) In the traffic management information, each node ID and a “travel permitted vehicle” representing the unmanned vehicle to which a travel permission regarding a travel zone (a travel zone to the next node on a route) represented by the node ID is given are associated with each other. As a travel permitted zone for the unmanned vehicle **20**, the traffic management control section **312** sets a zone ahead where the unmanned vehicle **20** can be permitted to travel, according to the position of the unmanned vehicle **20**. The unmanned vehicle **20** travels according to the node for the set zone.

(81) In the present embodiment, the travel permitted zone control scheme is adopted. Accordingly, in a case where a travel zone ahead of a travel permitted zone which has been set for a certain unmanned vehicle **20-1** is set as a travel permitted zone for another unmanned vehicle **20-2**, the traffic management control section **312** does not give, to the unmanned vehicle **20-1**, a travel permission regarding the travel zone ahead. In this case, the unmanned vehicle **20-1** stops in such a manner as not to pass the end node of the travel permitted zone which is currently permitted, and queues until a travel permission regarding the zone ahead is given.

(82) In addition, the traffic management control section **312** generates map information on the basis of work points specified in a work area, and causes the generated map information to be stored on the map information storage section **353**.

(83) FIG. 8 is a figure depicting an example of a travel route set for a work area where loading work is performed. As depicted in FIG. 8, in a case where a loading position (LP: Loading Point) (node_LP) is specified, the traffic management control section **312** generates a travel route **60** in the work area. The travel route **60** includes a plurality of travel zones including travel zones S1, S2, and S3.

(84) For example, the loading position (node_LP) is specified by the traffic management control section **312** on the basis of positional information regarding a loading machine **10** or the like. In addition, according to a positional relationship with the loading position, a queuing position (node_RP) where the following unmanned vehicle **20-2** queues in a case where the other unmanned vehicle **20-1** is positioned at the loading position is specified by the traffic management control

section **312**. Note that, in the case illustrated in the present embodiment, the loading position and the queuing position are set as turn around positions where the unmanned vehicles **20** turn around to switch to forward movement or backward movement.

(85) When the traffic management control section **312** specifies the loading position (node_LP) and the queuing position (node_RP), the traffic management control section **312** generates map information for allowing the unmanned vehicles **20** to travel, on the basis of undepicted nodes set at the entrance and exit of the work area, the queuing position (node_RP), and the loading position (node_LP).

(86) In the travel route **60** depicted in FIG. **8**, the travel zone S1 including the queuing position is the queuing zone S1 where the unmanned vehicle **20-2** heading toward the loading position queues, and includes a queuing position RP set as a turn around position. In addition, the travel zones S2 and S3 are a loading zone from the queuing zone to the loading position and an exit zone for the unmanned vehicle **20-1** to exit the loading position, respectively, and include the loading position LP set as a turn around position between the loading zone S2 and the exit zone S3. That is, the loading zone S2 and the exit zone S3 overlap at the loading position, and a common travel permission is given regarding the loading zone S2 and the exit zone S3, in the case considered here. That is, only one unmanned vehicle can enter the travel zones S2 and S3 at once. Accordingly, unless particular distinctions are made, the loading zone S2 is assumed to include the exit zone S3.

(87) Note that there are various possible map information generation methods. For example, map generation is performed by searching for an appropriate route on the basis of an index such as shortest route length from candidates of combinations of straight lines or arcs which are partial elements of a travel route, within an area in the work area in which generation of a travel route is tolerated.

(88) In addition, as for a work area where soil-dropping work is performed, the traffic management control section **312** generates map information similarly on the basis of a specified soil-dropping position. In that case, the soil-dropping position may be specified by an operator of a bulldozer or the like performing work in the work area or an operator who performs remote operation at the traffic management station **30**.

(89) In a case where the substitute vehicle travel-start time calculating section **313** has received a travel permission request regarding the queuing zone S2 from the unmanned vehicle **20-2** stopped and queuing at the queuing position of the queuing zone S1, the substitute vehicle travel-start time calculating section **313** outputs a travel permission regarding the queuing zone S1 to the unmanned vehicle **20-2** by a procedure like the one described below.

(90) First, the substitute vehicle travel-start time calculating section **313** calculates a time length taken for the unmanned vehicle **20-2** (a first unmanned vehicle) arriving at a deceleration start position where the unmanned vehicle **20-2** should start decelerating for stopping before entering the loading zone S2 after starting traveling from the queuing position.

(91) In addition, the substitute vehicle travel-start time calculating section **313** calculates a time length for arriving at a loading zone releasing position which is a time length taken for the unmanned vehicle **20-1** (a second unmanned vehicle) positioned in the loading zone S2 going out of the exit zone S3 after starting traveling from the loading position in response to reception of a loading completion notification to the unmanned vehicle **20-1**.

(92) Subsequently, the substitute vehicle travel-start time calculating section **313** calculates, on the basis of the time length taken for arriving at the deceleration start position and the time length for arriving at the loading zone releasing position, such a travel start time difference between the unmanned vehicle **20-1** and the unmanned vehicle **20-2** that the unmanned vehicle **20-1** goes out of the exit zone S3 a predetermined spare time length before the unmanned vehicle **20-2** arrives at the deceleration start position.

(93) Then, the substitute vehicle travel-start time calculating section **313** calculates a travel start time of the unmanned vehicle **20-2** on the basis of a reception time of the loading completion

notification to the unmanned vehicle **20-1** and the travel start time difference, and, after the travel start time has come, outputs a travel permission regarding the queuing zone S1 for the unmanned vehicle **20-2** (which can be said to include a travel start permission to start traveling from the queuing position here).

(94) Note that the travel permission request regarding the loading zone S2 is also issued from the following unmanned vehicle **20-2** simultaneously with a travel permission request regarding the queuing zone S1, and, simultaneously with the timing at which the preceding unmanned vehicle **20-1** goes out of the exit zone S3, a travel permission regarding the loading zone S2 for the unmanned vehicle **20-2** (which can be said to include an entry permission to allow entry into the loading zone S2 from the queuing zone S1 here) is output.

(95) Hereinbelow, details of a process performed by the traffic management controller **310** are explained in detail.

(96) FIG. **9** is a flowchart depicting the processing content of a process of giving permission of travel start, the process being performed by the traffic management controller.

(97) In FIG. **9**, when the traffic management controller **310** receives a travel permission request regarding the queuing zone S1 from the unmanned vehicle **20-2** stopped at the queuing position of the queuing zone S1 (Step **S1001**), the traffic management controller **310** assesses whether or not there is the preceding unmanned vehicle **20-1** in the loading zone S2 (Step **S1002**). In a case where the assessment result is NO, that is, in a case where there is not the preceding unmanned vehicle **20-1** in the loading zone, the traffic management controller **310** transmits a travel permission regarding the queuing zone S1 to the unmanned vehicle **20-2** (Step **S1007**), and ends the process.

(98) In addition, in a case where the result of the assessment in Step **S1002** is YES, that is, in a case where there is the preceding unmanned vehicle **20-1** in the loading zone S2, the substitute vehicle travel-start time calculating section **313** implements an estimation process of estimating a time length which is taken for the preceding unmanned vehicle **20-1** going out of the exit zone S3 and after which the loading zone S2 is released (Step **S1003**), implements an estimation process of estimating a time length taken for the unmanned vehicle **20-2** arriving at a deceleration start position where the unmanned vehicle **20-2** should start decelerating for stopping before entering the loading zone S2 after starting traveling from the queuing position (Step **S1004**), and implements a travel start time determination process of determining a travel start time of the unmanned vehicle **20-2** on the basis of a reception time of the loading completion notification to the unmanned vehicle **20-1** and a travel start time difference (Step **S1005**).

(99) Subsequently, the traffic management controller **310** assesses whether or not the travel start time calculated in Step **S1005** has come (Step **S1006**), and, in a case where the assessment result is NO, repeats the process in Step **S1006** until the assessment result becomes YES, that is, until the travel start time has come. In addition, in a case where the assessment result in Step **S1006** is YES, that is, in a case where the travel start time has come, a travel permission regarding the queuing zone S1 is transmitted to the unmanned vehicle **20-2** (Step **S1007**), and the process is ended.

(100) FIG. **10** is a flowchart depicting the processing content of the estimation process of estimating the time length for releasing a loading zone. In addition, FIG. **11** is a figure for explaining a processing principle of the estimation process of estimating the time length for releasing the loading zone.

(101) As depicted in FIG. **10** and FIG. **11**, in the estimation process of estimating the time length for releasing the loading zone, a loading zone releasing position x_{rel} is calculated (Step **S1201**), positions x ($k=1, \dots, N$) at intervals of Δx before the loading zone releasing position x_{rel} is reached are generated (Step **S1202**), a velocity v_k at each position x_k is calculated (Step **S1203**), a time length for travel between each pair of the division positions x_k is calculated from an average velocity (Step **S1204**), a time length T_{rel} for arriving at the loading zone releasing position, that is, a time length for releasing the loading zone, is calculated (Step **S1205**), and the process is ended.

(102) Note that, assuming that the acceleration (which is assumed to be constant) is α , a delay at

the time of starting traveling from LP is TL_{Pdel} , a travel permission releasing margin is L_{rel} , and the velocity limit is $v_{lim}(x)$, each numerical value is calculated in the following manner. Loading zone releasing position: $x_{rel}=d+L_{rel}$ Each of positions at intervals of distance Δx : $k\Delta x$ ($k=0, \dots, N$, $N=x_{rel}/\Delta x$ (rounded up)) Velocity at each x_k : $v_k=\min(\sqrt{2\alpha x_k}, v_{lim}(x_k))$ Time length for arriving at x_{rel} :

$$(103) T_{rel} = TL_{Pdel} + \sum_{k=0}^{N-1} (2 \Delta x / (v_k + v_{k+1}))$$

(104) FIG. 12 is a flowchart depicting the processing content of the estimation process of estimating a time length for arriving at the deceleration start position. In addition, FIG. 13 is a figure for explaining a processing principle of the estimation process of estimating the time length for arriving at the deceleration start position.

(105) As depicted in FIG. 12 and FIG. 13, in the estimation process of estimating the time length for arriving at the deceleration start position, a deceleration start position x_{dec} is calculated (Step S1401), positions x_k ($k=1, \dots, N$) at intervals of Δx before the deceleration start position x_{dec} are generated (Step S1402), the velocity v_k at each position x_k is calculated (Step S1403), a time length for travel between each pair of the division positions x_k is calculated from an average velocity (Step S1404), a time length T_{dec} for arriving at the deceleration start position is calculated (Step S1405), and the process is ended.

(106) Note that, assuming that the acceleration (which is assumed to be constant) is α , a delay at the time of starting traveling from RP is TR_{Pdel} , the loading zone entry velocity is v_0 , the deceleration (which is assumed to be constant) is β , a stop margin is L_{margin} , and the velocity limit is $v_{lim}(x)$, each numerical value is calculated in the following manner. Deceleration start position: $x_{dec}=d-(v_0^2/2\beta+L_{margin})$ Each of positions at intervals of distance Δx : $k\Delta x$ ($k=0, \dots, N$, $N=x_{rel}/\Delta x$ (rounded up)) Velocity at each x_k : $v_k=\min(\sqrt{2\alpha x_k}, v_{lim}(x_k))$ Time length for arriving at x_{dec} :

$$(107) T_{dec} = TR_{Pdel} + \sum_{k=0}^{N-1} (2 \Delta x / (v_k + v_{k+1}))$$

(108) Note that, as the time length for the releasing loading zone and the time length for arriving at the deceleration start position, for example, in cases such as a case where the same travel route is traveled repeatedly, ones that are calculated in advance can also be used by acquiring them as fixed values.

(109) FIG. 14 is a flowchart depicting the processing content of the travel start time determination process. In addition, FIG. 15 to FIG. 17 are figures for explaining a processing principle of the travel start time determination process.

(110) As depicted in FIG. 14 to FIG. 17, in the travel start time determination process, a travel start time difference is calculated (Step S1501), and it is assessed whether or not the travel start time difference is greater than 0 (zero) (Step S1502).

(111) In a case where the result of the assessment in Step S1502 is YES, that is, in a case where the travel start time difference is positive, it is assessed whether or not the preceding unmanned vehicle has started traveling (Step S1503). In a case where the assessment result is NO, the process in Step S1503 is repeated until the preceding unmanned vehicle starts traveling, that is, until the assessment result becomes YES. In addition, in a case where the result of the assessment in Step S1503 is YES, an optimum travel start time is determined on the basis of a travel start time (Step S1504), and the process is ended.

(112) In addition, in a case where the result of the assessment in Step S1502 is NO, that is, in a case where the travel start time difference is negative, it is assessed whether or not loading actions have been performed a predetermined number of times for the preceding unmanned vehicle (Step S1505), and, in a case where the assessment result is NO, the process in Step S1505 is repeated until the assessment result becomes YES. In addition, in a case where the result of the assessment in Step S1505 is YES, that is, in a case where loading actions have been performed a predetermined number of times, a travel start time of the preceding unmanned vehicle is estimated (Step S1506), an optimum travel start time is determined on the basis of the estimated travel start time (Step

S1504), and the process is ended.

(113) Note that, in a case where the travel start time difference is positive, assuming that, as depicted in FIG. 15, a travel start time of the preceding vehicle is t_0 , the time length for arriving at the loading zone releasing position is T_{rel} , the time length for arriving at the deceleration start position is T_{dec} , the travel start time difference is T_{diff} , the loading zone releasing time is $t_A = t_0 + T_{rel}$, and the deceleration start time is $t_B = t_0 + T_{diff} + T_{dec}$, $t_0 + T_{rel} < t_B = t_0 + T_{diff} + T_{dec}$ holds true from $t_A < t_B$, which is a condition that the travel start time difference should satisfy, and $T_{diff} > T_{rel} - T_{dec}$ is derived. Here, if a margin time length T_m is defined, a travel start time difference T_{diff} is determined according to $T_{rel} - T_{dec} + T_m$. In addition, in a case where the travel start time difference is negative, as depicted in FIG. 16, the travel start time difference T_{diff} can be determined according to $T_{rel} - T_{dec} + T_m$ by using an estimated time t_{01} as the travel start time of the preceding vehicle.

(114) In addition, as depicted in FIG. 17, a travel start time estimated in a case where the travel start time difference is negative can be determined by calculating the number of times of loading on the basis of vehicle size classes (each capacity is prestored) of a hydraulic excavator (the loading machine 10) and a dump truck (the unmanned vehicle 20). Specifically, the required number of times of loading N_L is determined according to (truck loading capacity)/(excavator bucket capacity) (rounded up), a final loading start time (a loading completion time of $(N_L - 1)$ -th loading) and a predicted time length for the loading completion notification (a time length from the final loading start time to a loading completion notification time) are defined, and an estimated loading completion notification time t_{01} is thereby calculated according to (final loading start time) + (predicted time length for the loading completion notification). The predicted time length for loading completion notification is learned on the basis of data of actual final loading start times and loading completion notification times, and a preset initial value is updated with the moving average value of recorded values. Note that the predicted time length for the loading completion notification may be learned independently for each combination of an operator and the vehicle size class of a hydraulic excavator.

(115) Advantages of the present embodiment configured as described above are explained in comparison with a conventional technology.

(116) FIG. 20 is a figure depicting a state of vehicle management in the conventional technology depicted as a comparative example, and FIG. 21 is a figure depicting changes of the position and velocity of an unmanned vehicle.

(117) As depicted in FIG. 20 and FIG. 21, in the conventional technology, in a case where a preceding unmanned vehicle in the loading zone S2 and a following unmanned vehicle stopped at the queuing position in the queuing zone S1 start traveling simultaneously (state a), the following unmanned vehicle arrives at the boundary between the queuing zone S1 and the loading zone S2 undesirably before the preceding unmanned vehicle goes out of the exit zone S3 (i.e., before a travel permission regarding the loading zone S2 for the following unmanned vehicle is acquired), and accordingly it becomes necessary for the following unmanned vehicle to pause in the queuing zone S1 until the a travel permission regarding the loading zone S2 is obtained (state b). Then, when the preceding unmanned vehicle goes out of the exit zone S3, the following unmanned vehicle acquires a travel permission regarding the loading zone S2 (state c), enters the loading zone S2 from the queuing zone S1 while accelerating (state d), and decelerates promptly to arrive at the loading position (state e). That is, in the conventional technology, since the frequency of acceleration and deceleration of the following unmanned vehicle increases, the average velocity lowers, and the work efficiency deteriorates. Also, a time length required for replacement of unmanned vehicles at the loading position increases, and the productivity deteriorates undesirably.

(118) FIG. 18 is a figure depicting a state of vehicle management in the present embodiment, and FIG. 19 is a figure depicting changes of the position and velocity of an unmanned vehicle.

(119) As depicted in FIG. 18 and FIG. 19, in the present embodiment, in a state where a preceding

unmanned vehicle in the loading zone S2 starts traveling from the loading position and a following unmanned vehicle keeps stopping at the queuing position in the queuing zone S1 (state a), according to a travel permission request regarding the queuing zone S1 from the following unmanned vehicle, a travel permission regarding the queuing zone S1 is issued after time corresponding to a time difference has elapsed from travel start of the preceding unmanned vehicle, and the following unmanned vehicle starts traveling (state b). At this time, the time difference between the travel start of the preceding unmanned vehicle and the travel start of the following unmanned vehicle is set optimally by the vehicle management system in the present embodiment, and the preceding unmanned vehicle goes out of the exit zone S3 while the following unmanned vehicle is traveling in the queuing zone S1 and before the following unmanned vehicle arrives at the deceleration start position. That is, in other words, the following unmanned vehicle acquires the travel permission regarding the loading zone S2 while traveling in the queuing zone S1 and before arriving at the deceleration start position (state c). Then, as a result, the following unmanned vehicle can enter the loading zone S2 and arrive at the loading position in a shorter time length from the issuance of the travel permission regarding the loading zone S2 and at the maximum velocity (velocity limit) without decelerating in the queuing zone S1 (state d, state e). Accordingly, a time length required for replacement of unmanned vehicles at the loading position can be shortened, and the productivity can be enhanced.

Second Embodiment

(120) A second embodiment of the present invention is explained with reference to FIG. 22 and FIG. 23. Note that constituent elements and the like in the present embodiment that are similar to those in the first embodiment are given the same reference characters, and explanations thereof are omitted as appropriate.

(121) In the case depicted in the present embodiment, such an optimum traveling velocity (velocity limit) that a time length for travel in the loading zone is minimized is applied to a following unmanned vehicle.

(122) FIG. 22 is a flowchart depicting the processing content of the process of giving permission of travel start, the process being performed by the traffic management controller according to the present embodiment. In addition, FIG. 23 is a figure for explaining a calculation principle of the optimum traveling velocity in the present embodiment.

(123) In FIG. 22, when the traffic management controller 310 receives a travel permission request regarding the queuing zone S1 from the unmanned vehicle 20-2 stopped at the queuing position of the queuing zone S1 (Step S1001), the traffic management controller 310 assesses whether or not there is the preceding unmanned vehicle 20-1 in the loading zone S2 (Step S1002). In a case where the assessment result is NO, that is, in a case where there is not the preceding unmanned vehicle 20-1 in the loading zone, the traffic management controller 310 transmits a travel permission regarding the queuing zone S1 to the unmanned vehicle 20-2 (Step S1007), and ends the process.

(124) In addition, in a case where the result of the assessment in Step S1002 is YES, that is, in a case where there is the preceding unmanned vehicle 20-1 in the loading zone S2, the substitute vehicle travel-start time calculating section 313 implements an optimum traveling velocity determination process of determining an optimum traveling velocity (optimum traveling velocity) of the following unmanned vehicle 20-2 on the basis of a distance across (length of) the loading zone S2 (Step S1901), implements the estimation process of estimating a time length taken for the preceding unmanned vehicle 20-1 going out of the exit zone S3 and in which the loading zone S2 is released (Step S1003), implements the estimation process of estimating a time length taken for the unmanned vehicle 20-2 arriving at a deceleration start position where the unmanned vehicle 20-2 should start decelerating for stopping before entering the loading zone S2 after starting traveling from the queuing position (Step S1004), and implements the travel start time determination process of determining a travel start time of the unmanned vehicle 20-2 on the basis of a reception time of the loading completion notification to the unmanned vehicle 20-1 and a travel start time difference

(Step S1005).

(125) Subsequently, the traffic management controller 310 assesses whether or not the travel start time calculated in Step S1005 has come (Step S1006), and, in a case where the assessment result is NO, repeats the process in Step S1006 until the assessment result becomes YES, that is, until the travel start time has come. In addition, in a case where the assessment result in Step S1006 is YES, that is, in a case where the travel start time has come, the optimum traveling velocity is applied as the velocity limit of the subject unmanned vehicle 20 (Step S1902), a travel permission regarding the queuing zone S1 is transmitted to the unmanned vehicle 20-2 (Step S1007), and the process is ended.

(126) As depicted in FIG. 23, there is a trade-off relationship between a traveling velocity v (velocity limit) of the following unmanned vehicle 20-2 and a time length T for travel taken until the unmanned vehicle 20-2 traveling in the loading zone S2 arrives at the loading position LP, which is a stop position, from the deceleration start position. This is because, as the traveling velocity v of the unmanned vehicle 20-2 increases, a distance L_{stop} from the deceleration start position (the distance from the deceleration start position to the loading position LP) increases, and the traveled distance required for the unmanned vehicle 20-2 to decelerate and stop increases. On the other hand, as depicted in FIG. 23, there is a traveling velocity (optimum traveling velocity) of an unmanned vehicle that minimizes the time length T for travel.

(127) Here, assuming that the velocity of entry of the following unmanned vehicle 20-2 from the queuing zone S1 to the loading zone S2 is v_0 , the deceleration from the deceleration start position to the loading position is β , a stop margin is L_{margin} , and the distance across (length of) the loading zone S1 is LS_2 , the optimum traveling velocity is determined in the following manner. Distance from deceleration start position to loading position (distance from deceleration start position): $L(v_0)$

(128) $T(v_0) = 2 \sqrt{((L_{margin} + LS_2) / \beta)}$ time length for travel at distance from deceleration start position: $T(v_0)$

(129) $L(v_0) = v_0^2 / 2\beta + L_{margin} + LS_2$ Optimum traveling velocity: v_{01}

$v_{01} = \sqrt{\beta(L_{margin} + LS_2)}$ (from the inflection point of $T(v_0)$) Shortest time length for travel: v_{01}

(130) $T(v_0) = L(v_0) / v_0 + v_0 / 2\beta = v_0 / \beta + (L_{margin} + LS_2) / v_0$

(131) In other respects, the configuration is similar to that of the first embodiment.

(132) Also in the present embodiment configured as described above, advantages similar to those in the first embodiment can be attained.

(133) In addition, by applying the optimum traveling velocity as the velocity limit, it is possible to further shorten a time length required for replacement of unmanned vehicles at the loading position and enhance productivity.

Modification Example of Second Embodiment

(134) A modification example of the second embodiment of the present invention is explained with reference to FIG. 24. Note that constituent elements and the like in the present modification example that are similar to those in the second embodiment are given the same reference characters, and explanations thereof are omitted as appropriate.

(135) In the case depicted in the present embodiment, such an optimum traveling velocity (velocity limit) that the fuel consumption in the loading zone is minimized is applied to a following unmanned vehicle.

(136) FIG. 24 is a figure for explaining a calculation principle of the optimum traveling velocity according to the present modification example.

(137) As depicted in FIG. 24, there is a trade-off relationship between the traveling velocity v (velocity limit) of the following unmanned vehicle 20-2 and fuel consumption F consumed until the unmanned vehicle 20-2 traveling in the loading zone S2 arrives at the loading position LP, which is a stop position, from the deceleration start position. This is because, as the traveling velocity v of the unmanned vehicle 20-2 increases, the unmanned vehicle 20-2 can travel in a

shorter time length but the fuel consumption worsens and, as the traveling velocity v decreases, the fuel consumption is improved but the time length for travel of the unmanned vehicle **20-2** becomes longer. On the other hand, as depicted in FIG. **24**, there is a traveling velocity (optimum traveling velocity) of an unmanned vehicle that minimizes the fuel consumption F . In the present modification example, this optimum traveling velocity is applied as the velocity limit of the following unmanned vehicle **20-2**.

(138) In other respects, the configuration is similar to that of the second embodiment.

(139) Also in the present modification example configured as described above, advantages similar to those in the second embodiment can be attained.

(140) In addition, by applying the optimum traveling velocity as the velocity limit, the fuel consumption can be enhanced.

(141) Note that, whereas the optimum traveling velocity is calculated on the basis of the relationship between the traveling velocity and the fuel consumption in the present modification example, this is not the sole example. For example, using an evaluation function in which a function $T(v_0)$ of a time length for travel related to the velocity v_0 and a function $F(v_0)$ of the fuel consumption are given weights, such an optimum traveling velocity that the time length for travel and the fuel consumption are minimized may be calculated. In addition, in another manner of the configuration, such an optimum speed of rotation of the engine that the fuel consumption is minimized is calculated.

Third Embodiment

(142) A third embodiment of the present invention is explained with reference to FIG. **25**. Note that constituent elements and the like in the present embodiment that are similar to those in the first embodiment are given the same reference characters, and explanations thereof are omitted as appropriate.

(143) In the case depicted in the present embodiment, the time length for releasing the loading zone is updated according to the travel state of the preceding unmanned vehicle such as its position or velocity, and the travel start time difference T_{diff} is updated according to the updated time length for releasing the loading zone.

(144) FIG. **25** is a flowchart depicting the processing content of the process of giving permission of travel start, the process being performed by the traffic management controller according to the present embodiment.

(145) In FIG. **25**, when the traffic management controller **310** receives a travel permission request regarding the queuing zone $S1$ from the unmanned vehicle **20-2** stopped at the queuing position of the queuing zone $S1$ (Step **S1001**), the traffic management controller **310** assesses whether or not there is the preceding unmanned vehicle **20-1** in the loading zone $S2$ (Step **S1002**). In a case where the assessment result is NO, that is, in a case where there is not the preceding unmanned vehicle **20-1** in the loading zone, the traffic management controller **310** transmits a travel permission regarding the queuing zone $S1$ to the unmanned vehicle **20-2** (Step **S1007**), and ends the process.

(146) In addition, in a case where the result of the assessment in Step **S1002** is YES, that is, in a case where there is the preceding unmanned vehicle **20-1** in the loading zone $S2$, the substitute vehicle travel-start time calculating section **313** implements the estimation process of estimating a time length which is taken for the preceding unmanned vehicle **20-1** going out of the exit zone $S3$ and after which the loading zone $S2$ is released (Step **S1003**), implements the estimation process of estimating a time length taken for the unmanned vehicle **20-2** arriving at a deceleration start position where the unmanned vehicle **20-2** should start decelerating for stopping before entering the loading zone $S2$ after starting traveling from the queuing position (Step **S1004**), and implements the travel start time determination process of determining a travel start time of the unmanned vehicle **20-2** on the basis of a reception time of the loading completion notification to the unmanned vehicle **20-1** and a travel start time difference (Step **S1005**).

(147) Subsequently, the traffic management controller **310** assesses whether or not the travel start

time calculated in Step S1005 has come (Step S1006). In a case where the assessment result is NO, the traffic management controller 310 updates the time length Trel for releasing the loading zone, on the basis of the position or velocity of the preceding vehicle. In a case where the travel start time difference Tdiff calculated according to the updated time length Trel for releasing the loading zone is smaller than the previously-updated travel start time difference Tdiff, the traffic management controller 310 updates the optimum travel start time with the calculated value based on the travel start time difference Tdiff (Step S2001), and returns to the process in Step S1006.

(148) In addition, in a case where the assessment result in Step S1006 is YES, that is, in a case where the travel start time has come, the optimum traveling velocity is applied as the velocity limit of the subject unmanned vehicle 20 (Step S1902), and the process is ended.

(149) In other respects, the configuration is similar to that of the first embodiment.

(150) Also in the present embodiment configured as described above, advantages similar to those in the first embodiment can be attained.

(151) In addition, by updating the optimum travel start time such that it is optimized, it is possible to further shorten a time length required for replacement of unmanned vehicles at the loading position and enhance productivity.

Fourth Embodiment

(152) A fourth embodiment of the present invention is explained with reference to FIG. 26 and FIG. 27. Note that constituent elements and the like in the present embodiment that are similar to those in the first embodiment are given the same reference characters, and explanations thereof are omitted as appropriate.

(153) In the case depicted in the present embodiment, a travel permission (travel start permission) regarding the queuing zone for a following unmanned vehicle is issued on the basis of the position of a preceding unmanned vehicle in the loading zone.

(154) FIG. 26 is a figure for explaining a processing principle of travel permission output in the present embodiment.

(155) As depicted in FIG. 26, the following unmanned vehicle 20-2 in the queuing zone S1 starts traveling with a delay of the “travel start time difference” after the preceding unmanned vehicle 20-1 in the loading zone S2 has started traveling, and accordingly the position (target arrival position) at which the preceding unmanned vehicle 20-1 arrives after the “travel start time difference” can be identified. Then, by triggering travel start of the following unmanned vehicle 20-2 (i.e., by triggering issuance of a travel permission) upon the arrival of the preceding unmanned vehicle 20-1 at the target arrival position, actions similar to those in the first embodiment can be realized. That is, in the present embodiment, by using the position of the preceding unmanned vehicle as a trigger instead of a time length (time), actions similar to those in the first embodiment are realized.

(156) Here, assuming that the acceleration of the preceding unmanned vehicle 20-1 is αL and the maximum velocity of the preceding unmanned vehicle 20-1 is v_{max} , the target arrival position is calculated in the following manner according to $(\text{travel start time difference } T_{diff}) = T_{rel} - T_{dec} + T_m$.

(157) That is, in a case of (1) $T_{diff} \leq v_{max}/\alpha L$ in FIG. 26, the target arrival position is $\alpha L \times T_{diff}$ 2. In addition, in a case of (2) $T_{diff} > v_{max}/\alpha L$, the target arrival position is $v_{max} \times T_{diff} - v_{max}^2/2\alpha L$.

(158) FIG. 27 is a figure depicting a state of vehicle management according to the present embodiment.

(159) As depicted in FIG. 27, in the present embodiment, in a state where a preceding unmanned vehicle in the loading zone S2 starts traveling from the loading position and a following unmanned vehicle keeps stopping at the queuing position of the queuing zone S1 (state a), a travel permission regarding the queuing zone S1 is issued for the following unmanned vehicle when the preceding unmanned vehicle has arrived at the target arrival position in the exit zone S3, and the following unmanned vehicle starts traveling (state b). At this time, the time difference between the travel start of the preceding unmanned vehicle and the travel start of the following unmanned vehicle is set

such that the time difference is optimized by the vehicle management system as in the first embodiment, and the preceding unmanned vehicle goes out of the exit zone S3 while the following unmanned vehicle is traveling in the queuing zone S1 and before arriving at the deceleration start position. That is, in other words, the following unmanned vehicle acquires the travel permission regarding the loading zone S2 while traveling in the queuing zone S1 and before arriving at the deceleration start position (state c). Then, as a result, the following unmanned vehicle can enter the loading zone S2 and arrive at the loading position in a shorter time length from the issuance of the travel permission regarding the loading zone S2 and at the maximum velocity (velocity limit) without decelerating in the queuing zone S1 (state d, state e). Accordingly, a time length required for replacement of unmanned vehicles at the loading position can be shortened, and the productivity can be enhanced.

(160) In other respects, the configuration is similar to that of the first embodiment.

(161) Also in the present embodiment configured as described above, advantages similar to those in the first embodiment can be attained.

Fifth Embodiment

(162) A fifth embodiment of the present invention is explained with reference to FIG. 28. Note that constituent elements and the like in the present embodiment that are similar to those in the first embodiment are given the same reference characters, and explanations thereof are omitted as appropriate.

(163) Note that, in the case depicted in the present embodiment, the loading position and the queuing position are set not as turn around positions where unmanned vehicles 20 turn around to switch to forward movement or backward movement, but as stop passing positions where the unmanned vehicles 20 enter from one side and exit from the other side.

(164) FIG. 28 is a figure depicting an example of a travel route in the present embodiment.

(165) As depicted in FIG. 28, on the travel route 60, a travel zone so is a queuing zone S0 where the unmanned vehicle 20-2 heading toward the loading position queues, and, at an end section thereof, the queuing position (QP: Queuing Point) set as a stop passing position is arranged. In addition, the travel zone S1 is an approach zone S1 where the following unmanned vehicle 20-2 having started traveling from the queuing position QP in the queuing zone S0 travels when entering the loading exit zones S2 and S3. In addition, the travel zones S2 and S3 are the loading exit zones S2 and S3 combining features of both the loading zone S2 and the exit zone S3 in the first embodiment, and includes the loading position (LP: Loading Point) set as a stop passing position. A common travel permission is given regarding the loading exit zones S2 and S3, and only one unmanned vehicle can enter the loading exit zones S2 and S3 at once.

(166) In the present embodiment configured as described above, by treating the approach zone S1 and the loading exit zone S2 and S3 together similarly to the loading zone S2 and the exit zone S3 of the first embodiment, actions similar to those in the first embodiment can be performed.

(167) In other respects, the configuration is similar to that of the first embodiment.

(168) Also in the present embodiment configured as described above, advantages similar to those in the first embodiment can be attained.

(169) Note that, in the work area which is depicted in FIG. 28 and where the loading work is performed, the route from the entrance position to the loading position LP and the route from the loading route LP to the exit position may be considered as separate routes, the travel zone S2 and the travel zone S3 may be generated independently, and the loading position LP may be set at a position where the travel zone S2 and the travel zone S3 overlap.

Supplementary Notes

(170) Note that the present invention is not limited to the embodiments described above, and includes various modification examples and combinations within the scope not departing from the gist of the present invention. In addition, the present invention is not limited to those including all the constituent elements explained in the embodiments described above, and also includes those

from which some of the constituent elements are deleted.

(171) In addition, some or all of the constituent elements, functions, and the like described above may be realized by designing them, for example, in an integrated circuit. In addition, the constituent elements, functions, and the like described above may be realized by software by a processor interpreting and executing a program to realize respective functions.

DESCRIPTION OF REFERENCE CHARACTERS

(172) **10**: Loading machine (hydraulic excavator) **10A**: Front device **10B**: Upper swing structure **10C**: Lower travel structure **20**: Unmanned vehicle (dump truck) **30**: Traffic management station **40**: Wireless communication line **41**: Wireless base station **60**: Travel route (transportation route) **100**: Vehicle management system **140**: Wireless communication device **170**: Loading completion notification input device **171**: Vehicle body operation input device (operation lever) **172**: Operator's seat **180**: Vehicle body drive system **191**: Boom **192**: Arm **193**: Bucket **194**: Boom cylinder **195**: Arm cylinder **196**: Bucket cylinder **197**: Operation room **200**: Unmanned vehicle controller **201**: Autonomous travel control section **202**: Vehicle body information managing section **203**: Travel permission requesting section **210**: Travel drive system **220**: Position/orientation sensor **230**: Velocity sensor **240**: Wireless communication device **250**: Storage device **251**: Map information storage section **270**: Loading sensor **281**: Vehicle body frame **282L**, **282R**: Follower wheel **283L**, **283R**: Driving wheel **284**: Operation room **285**: Cargo bed (vessel) **285a**: Pin-joining section **286**: Hoist cylinder **289**: Fuel tank **290L**, **290R**: Travel motor **310**: Traffic management controller **311**: Vehicle allocation managing section **312**: Traffic management control section **313**: Substitute vehicle travel-start time calculating section **340**: Wireless communication device **350**: Traffic management storage device **351**: Vehicle allocation management information storage section **352**: Traffic management information storage section **353**: Map information storage section

Claims

1. A vehicle management system in which a plurality of unmanned vehicles including at least a first unmanned vehicle and a second unmanned vehicle that haul to-be-hauled objects and are capable of traveling autonomously, a loading machine that performs loading work of loading the to-be-hauled objects onto each of the plurality of unmanned vehicles, and a traffic management station that performs traffic management control of the loading machine and the plurality of unmanned vehicles are wirelessly connected to each other in a mutually communicable manner, the vehicle management system comprising: an unmanned vehicle controller that acquires vehicle body information including positional information representing positions of the plurality of unmanned vehicles at a work site and orientation information representing orientations of the plurality of unmanned vehicles at the work site, controls autonomous traveling of the plurality of unmanned vehicles on a basis of a travel permission for the plurality of unmanned vehicles, and outputs a travel permission request for requesting a travel permission, for each of the plurality of unmanned vehicles, regarding a predetermined travel zone among a plurality of travel zones included in a travel route where the plurality of unmanned vehicles travel; a loading completion notification input device that outputs, according to input operation by an operator of the loading machine, a loading completion notification indicating that loading work of loading the to-be-hauled objects onto corresponding one of the unmanned vehicles that is stopped at a loading position preset on the travel route as a position where the loading work is performed by the loading machine for the unmanned vehicles has been completed; and a traffic management controller that sets a travel route where the plurality of unmanned vehicles travel, on a basis of a work schedule predetermined for the work site and map information including information regarding positions and velocity limits of a plurality of travel routes and that outputs a travel permission regarding a predetermined travel zone of the travel route set for the plurality of unmanned vehicles, on a basis of a travel permission request from the unmanned vehicle controller, wherein the traffic management controller is

configured to, in a case where a loading zone which is a travel zone including the loading position and in which only one unmanned vehicle is allowed to enter at once and a queuing zone which is a travel zone adjacent to the loading zone and in which an unmanned vehicle heading toward the loading zone queues are set, and where a travel permission request regarding the queuing zone by the first unmanned vehicle stopped in the queuing zone is received, calculate such a travel start time difference between the first unmanned vehicle and the second unmanned vehicle positioned in the loading zone that the second unmanned vehicle goes out of the loading zone a predetermined spare time length before the first unmanned vehicle arrives at a deceleration start position where the first unmanned vehicle should start decelerating for stopping before entering the loading zone, on a basis of a time length for arriving at the deceleration start position that is a time length taken for the first unmanned vehicle arriving at the deceleration start position after starting traveling, and on a basis of a time length for arriving at a loading zone releasing position that is a time length taken for the second unmanned vehicle going out of the loading zone after starting traveling in response to reception of the loading completion notification to the second unmanned vehicle, calculate a travel start time of the first unmanned vehicle on a basis of a reception time of the loading completion notification to the second unmanned vehicle and the travel start time difference, and output a travel permission regarding the queuing zone for the first unmanned vehicle after the travel start time has come.

2. The vehicle management system according to claim 1, wherein each of the plurality of unmanned vehicles has the unmanned vehicle controller, the loading machine has the loading completion notification input device, and the traffic management station has the traffic management controller.

3. The vehicle management system according to claim 1, wherein the traffic management controller is configured to calculate the number of times of loading actions that is necessary until loading work for the second unmanned vehicle is completed, on a basis of a loaded amount of the to-be-hauled objects loaded onto the second unmanned vehicle at one loading action by the loading machine and a planned loaded amount of the second unmanned vehicle, estimate a reception time of the loading completion notification on a basis of the number of times of loading actions and a loading action start time in a case where the travel start time difference is negative, calculate a travel start time of the first unmanned vehicle on a basis of the estimated reception time of the loading completion notification and the travel start time difference, and output a travel permission regarding the queuing zone for the first unmanned vehicle after the travel start time has come.

4. The vehicle management system according to claim 3, wherein the traffic management controller is configured to estimate the reception time of the loading completion notification on a basis of a pre-learned time difference between a loading time and the reception time of the loading completion notification.

5. The vehicle management system according to claim 1, wherein the traffic management controller is configured to set a velocity of entry of the first unmanned vehicle into the loading zone.

6. The vehicle management system according to claim 5, wherein the traffic management controller is configured to set the velocity of entry of the first unmanned vehicle into the loading zone on a basis of a distance across the loading zone.

7. The vehicle management system according to claim 5, wherein the traffic management controller is configured to set the velocity of entry of the first unmanned vehicle into the loading zone such that a time length for travel in the loading zone is minimized.

8. The vehicle management system according to claim 1, wherein the traffic management controller is configured to, in a case where a loading zone which is a travel zone including the loading position and in which only one of the unmanned vehicles is allowed to enter at once and a queuing zone which is a travel zone adjacent to the loading zone and in which corresponding one of the unmanned vehicles heading toward the loading zone queues are set, and where a travel permission request regarding the queuing zone for the first unmanned vehicle stopped in the queuing zone is

received, calculate such a travel start time difference between the first unmanned vehicle and the second unmanned vehicle positioned in the loading zone that the second unmanned vehicle goes out of the loading zone a predetermined spare time length before the first unmanned vehicle arrives at a deceleration start position where the first unmanned vehicle should start decelerating for stopping before entering the loading zone, on a basis of a time length for arriving the deceleration start position that is a time length taken for the first unmanned vehicle arriving at the deceleration start position after starting traveling, and on a basis of a time length for arriving at a loading zone releasing position that is a time length taken for the second unmanned vehicle going out of the loading zone after starting traveling in response to reception of the loading completion notification to the second unmanned vehicle, calculate a target arrival position where the second unmanned vehicle arrives after the travel start time difference, and output a travel permission regarding the queuing zone for the first unmanned vehicle when the second unmanned vehicle arrives at the target arrival position.
